Math 514, F19
Quiz 9

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Consider an experiment with four possible outcomes and suppose that the quoted odds for the first three outcomes are as follows.

Outcome	1	2	3
Odds	2	3	4

What must be the odds againt outcome 4 to avoid an arbitrage?

We need to see whether there exists a probability vector (P_1, P_2, P_3, P_4) for which all bets are fair. Thus we have.

$$O_i P_i + (-1)(1-P_i) = 0 \quad \text{for } i=1,2,3,4$$

$$\Rightarrow \quad P_i = \frac{1}{1+O_i}$$

At the same time, we know $\sum_{i=1}^{4} P_i = 1$, thus

$$\frac{1}{1+O_1} + \frac{1}{1+O_2} + \frac{1}{1+O_3} + \frac{1}{1+O_4} = 1$$

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{1+O_4} = 1 \Rightarrow \frac{47}{60} + \frac{1}{1+O_4} = 1 \Rightarrow 1+O_4 = \frac{60}{13} \Rightarrow O_4 = \frac{47}{13}$$

Problem 2. (5 points) An experiment can result in any of the outcomes 1,2, or 3. If there are two different wagers with

$$r_1(1) = 2,\ r_1(2) = 4,\ r_1(3) = -3,$$
$$r_2(1) = 4,\ r_2(2) = 6,\ r_2(3) = -5,$$

Is an arbitrage possible?

If an arbitrage is impossible, then there is a probability vector (P_1, P_2, P_3) such that

$$\begin{cases} \sum_{j=1}^{3} P_j r_1(j) = 0 \\ \sum_{j=1}^{3} P_j r_2(j) = 0 \\ \sum_{j=1}^{3} P_j = 1 \end{cases} \Rightarrow \begin{cases} 2P_1 + 4P_2 - 3P_3 = 0 \\ 4P_1 + 6P_2 - 5P_3 = 0 \\ P_1 + P_2 + P_3 = 1 \end{cases}$$

$$\Rightarrow \begin{cases} 5P_1 + 7P_2 = 3 \\ 9P_1 + 11P_2 = 5 \end{cases} \Rightarrow \begin{cases} P_1 = \frac{1}{4} \\ P_2 = \frac{1}{4} \end{cases}$$, thus there is no arbitrage.

$$\Rightarrow P_3 = \frac{1}{2}$$